# Leon Noble

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## Summary

**Research Interests:** Cosmic Dawn and Epoch of Reionization, post-reionization Universe, 21 cm Cosmology, Line Intensity Mapping, statistics of non-Gaussian signal, simulations of CD-EoR and Large Scale Structures, Simulation Based Inference for Line Intensity Mapping

**Major science contributions:**

→ Modeling and forecasts for the detectability of the post-EoR 21 cm-galaxy cross-bispectrum for all unique triangles with SKA-Mid and Euclid-like galaxy survey
→ Developed a fast 21 cm-galaxy cross-bispectrum estimator for all unique triangles
→ Showed that 21 cm bispectrum observations with SKA-Low can distinguish between different reionization morphologies which arise due to changes in the source properties
→ Showed that joint inference with multi-line intensity mapping can give improved constraints on reionization sources
→ Developed GPU-accelerated simulation suite for Multi-line Intensity Mapping (21 cm, CII, and CO)
→ Developed a Simulation-Based Inference pipeline for Multi-line Intensity Mapping and their cross-correlations

**Publications in refereed journals:** 2 as first author, 1 as second author, 5 as co-author

**Contribution to new SKAO science white book (AASKAII) chapters:** Contributed to 4 chapters

**Leadership:** Co-lead the HIstorian team in SKA Data Challenge 3b: Inference, we were placed in the leaderboard

**Talks and Presentations:** 16 talks and 5 poster presentations in various international conferences

**Fellowships and Awards:** Recipient of DST-INSPIRE fellowship, ICTP-STEP fellowship, Best Speaker Award in the Young Astronomer Meet 2025

## Education

**Ph.D. in Astronomy** July 2022 – Expected July 2027
*Department of Astronomy, Astrophysics and Space Engineering*
*Indian Institute of Technology Indore (IIT Indore), India*

- Thesis title: Probing the High-Redshift Universe Using Multi-Line Intensity Mapping and Cross-Correlations
- Supervisor: Dr. Suman Majumdar

**M.Sc. Physics (CGPA: 9/10)** Aug 2019 – Sep 2021
*National Institute of Technology Karnataka (NITK), Mangaluru, India*

- Thesis: "Numerical Investigation of Inflationary Scenario", supervised by Dr. V. Sreenath

**B.Sc. Physics (CGPA: 9.6/10)** Jun 2016 – May 2019
*Sacred Heart College Thevara, Kochi, Kerala, India*

## Publications

**2 first-author, 1 second-author (under review), 5 co-authored, and 4 SKAO AASKA II chapters.** Complete list in Appendix A (p. 3).

**INSPIRE-HEP:** https://inspirehep.net/authors/2967305?ui-citation-summary=true
**Google Scholar:** https://scholar.google.com/citations?user=LSritLYAAAAJ&hl=en
**arXiv:** https://arxiv.org/a/noble_l_1.html
**ORCiD:** https://orcid.org/0009-0004-3138-1130

## Presentations

**16 oral presentations and 5 poster presentations in various international conferences.** Full list in Appendix B (p. 4).

## Fellowships & Awards

**ICTP STEP Fellowship** 2024 – 2026
*To visit ICTP and SISSA, and collaborate with Prof. Matteo Viel (SISSA) and researchers in the Trieste area*

**DST INSPIRE Fellowship, Department of Science and Technology, Government of India** Feb 2023 – Present

*Covers PhD salary and a research grant*

**Best Speaker Award, Young Astronomer's Meet 2025, IIT (BHU) Varanasi, India** — Dec 2025

**DST INSPIRE Scholarship, Department of Science and Technology, Government of India** — 2016 – 2021

*For a constant 5 year contract during B.Sc and M.Sc (80,000 INR per annum)*

## Travel Grants

- **Simons Foundation** — To attend "New Frontiers in Cosmology 2026", A Coruña, Spain *(Aug 2026)*
- **SKA-India Consortium (SKAIC)** — To attend "A New Era in Astrophysics: Preparing for Early Science with the SKAO", Görlitz, Germany *(Jun 2025)*
- **ICTP** — To attend "ICTP Summer School on Cosmology", Trieste, Italy *(Jun 2024)*

## Research Visits

**International Centre for Theoretical Physics (ICTP) & Scuola Internazionale Superiore di Studi Avanzati (SISSA)** | Trieste, Italy

31 Aug 2026 – 17 Nov 2026
15 May 2025 – 10 Aug 2025
28 Oct 2024 – 19 Dec 2024
17 Jun 2024 – 5 Jul 2024

- Visiting researcher with Prof. Matteo Viel. Four extended visits between Jul 2024 – Nov 2026 (≈9 months total) as part of ICTP STEP fellowship

**Scuola Normale Superiore (SNS)** | Pisa, Italy — 8 – 12 Jul 2024

- Research collaborator with Prof. Andrei Mesinger

## Leadership & Collaboration

- **Co-lead, HIstorian team** — SKA Data Challenge 3b: Inference, we were placed in the leaderboard
- **Associate member**, SKA Cosmic Dawn-Epoch of Reionization Science Working Group
- **Associate member**, SKA Cosmology Science Working Group
- **Co-author of four SKAO science white book (AASKAII) chapters,** *Advancing Astrophysics with the Square Kilometre Array II* (AASKA II), 2026

## Technical Skills

**Computing time awarded:** 10k CPU-hours in Pegasus (IUCAA) (**PI**, 2024), 10k CPU-hours in CSCS Alps (**Co-PI**, 2024–25)

**Simulations Developed:**

- Cross-bispectrum estimator for all unique triangles (used in arXiv:2604.15440)
- GPU-accelerated multi-line intensity mapping simulations (21 cm, CII, CO)
- SBI pipeline for multi-line intensity mapping and their cross-correlations

**Cosmological simulations used:** ReionYuga, 21cmFAST; Particle-Mesh N-body, FastPM; LIM simulator

**Inference & cosmology packages used:** polychord, PyMultiNest, Cobaya, emcee, CosmoMC, CAMB

**HPC facilities used:** CSCS Alps, Pegasus (IUCAA), Jamadagni (IIT Indore), NVIDIA DGX A100 Server (IIT Indore)

**Languages & tools:** Python, C++, C; Git; Docker, Singularity

## Teaching

- Teaching Assistant, tutorials on *Hands-on with Reionization Simulations & Statistical Inference from Cosmological Signal*, **School on Radio Cosmology and Continuum Observations in the SKA Era**, ICTS Bengaluru, India, 14–18 Apr 2025
- Teaching Assistant, *AA 471/671: Relativity and Cosmology*, Autumn 2024 & 2023
- Teaching Assistant, *AA 608: Astrostatistics*, Spring 2024 & 2025

### As First Author

1. **L. Noble**, S. Majumdar, M. Viel, F. Fontanot, G. De Lucia, A. K. Shaw, M. Spinelli, M. Kamran, L. Xie, M. Hirschmann, ***Probing the large-scale structure with 21cm-galaxy cross-bispectrum: Estimates from simulations and forecasts for upcoming cosmological surveys***, 2026, ApJ, 1005, 218
2. **L. Noble**, M. Kamran, S. Majumdar, C. S. Murmu, R. Ghara, G. Mellema, I. T. Iliev, J. R. Pritchard, ***Impact of the sources of Epoch of Reionization on the 21-cm bispectrum***, 2024, JCAP10(2024)003
3. **L. Noble**, Y. Mahida,S. Majumdar, M. M. Dosibhatla, ***21-cm–[C II] cross-correlations: improving constraints on EoR source parameters with the SKAO in a simulation-based inference framework*** (in prep., to be submitted to *ApJ*)

### As Second Author

4. V. Pandya, **L. Noble**, S. Majumdar, D. Sarkar, M. Kamran, A. Datta, ***Probing Reionization up to the Mark: The Marked Power Spectrum to unveil the HI 21-cm signal from the EoR*** (under review, JCAP), arXiv:2605.18301

### With Significant Contribution

5. I. Nasreen, K. K. Datta, A. K. Shaw, **L. Noble**, R. Ghara, Sk. S. Ali, A. Mishra, M. Kamran, S. Majumdar, ***The Effect of Large Optical Depths on the Non-Gaussian 21-cm signal from Cosmic Dawn***, 2025, MNRAS, 543, 2553
6. J. Wasserman, E. Zackrisson, J. Dhandha, A. Fialkov, **L. Noble**, S. Majumdar, ***Ultraviolet photon production rates of the first stars: Impact on the He* ii *λ1640 Å emission line from primordial star clusters and the 21-cm signal from cosmic dawn***, 2026, MNRAS, 547, stag386
7. Y. Mahida, S. K. Yadav, S. Majumdar, **L. Noble**, C. S. Murmu, S. Dasgupta, S. Dutta, H. Tiwari, A. K. Shaw, ***From ANN to BNN: Inferring Reionization Parameters using Uncertainty-aware Emulators of 21-cm Summaries***, 2025, JCAP12(2025)055
8. P. R. Posture, Y. Mahida, S. Majumdar, **L. Noble**, ***CosmoUiT: A Vision Transformer-UNet Hybrid for Fast and Accurate Emulation of 21-cm Maps from the Epoch of Reionization***, 2026, JCAP06(2026)011
9. H. Potluri, M. M. Dosibhatla, **L. Noble**, C. S. Murmu, S. Majumdar, S. K. Pal, S. Dasgupta, S. Bag, A. Datta, ***Interpreting the HI 21-cm cosmology maps through Largest Cluster Statistics III: Impact of the lightcone effect*** (under review, JCAP), arXiv:2602.16651

### SKAO Chapters

**Co-author of four chapters of** *Advancing Astrophysics with the Square Kilometre Array II* **(AASKA II), 2026.**

10. ***Cosmology with Intensity Mapping via Statistics Beyond the Power Spectrum in the SKAO Era***
S. Majumdar, D. Sarkar, **L. Noble**, M. M. Dosibhatla, et al. AASKAII/Majumdar01, arXiv:2606.30200
11. ***Cosmology with Multi-Wavelength Line Intensity Mapping Synergies in the SKAO Era***
D. Sarkar, S. Majumdar, C. S. Murmu, **L. Noble**, et al. AASKAII/Sarkar01, arXiv:2606.30224
12. ***Inferring Cosmology and Astrophysics from the High-redshift 21cm Signal with SKA-Low***
A. Acharya, S. Bag, S. Berger, M. Bianco, et al., including **L. Noble**. AASKAII/Acharya01, arXiv:2606.27858
13. ***Overview of 21cm Experiments at High Redshift with SKAO***
G. Bernardi, D. Breitman, A. Datta, A. Fialkov, et al., including **L. Noble**. AASKAII/Trott01, arXiv:2606.26435

## CONFERENCE TALKS

1. *21cm-CII cross-correlations: Improving the constraints on EoR source parameters via SKA*, **Square Kilometer Array Epoch of Reionization Science Team meeting 2026**, Palata Nauke, Belgrade, Serbia. *7–11 Sep 2026*
2. *The 21cm-Galaxy Cross-Bispectrum: A Forecast for Synergies between SKA-Mid and Galaxy Surveys*, **Workshop on 21-cm Cosmology**, TIFR Mumbai, India. *9–11 Mar 2026*
3. *The 21cm-Galaxy Cross-Bispectrum: A Forecast for Synergies between SKA-Mid and Galaxy Surveys*, **Cosmology at Multiwavelength**, IIT Indore, India. *10–11 Feb 2026*
4. *SKAO-Euclid Synergies of the post-Reionization Universe: 21cm-Galaxy Cross-Bispectrum*, **Young Astronomers' Meet 2025**, IIT (BHU) Varanasi, India. *5–7 Dec 2025*
5. *SKAO-Euclid Synergies of the post-Reionization Universe: 21cm-Galaxy Cross-Bispectrum*, **A New Era in Astrophysics: Preparing for Early Science with the SKAO**, Görlitz, Germany. *16–20 Jun 2025*
6. *SKAO-Euclid Synergies of the post-Reionization Universe: 21cm-Galaxy Cross-Bispectrum*, **Line Intensity Mapping 2025**, LAPTh Annecy, France. *2–6 Jun 2025*
7. *Reionization and Beyond via Higher-Order Statistics of 21-cm Maps*, **Radio Cosmology and Continuum Observations in the SKA Era: A Synergic View**, ICTS Bengaluru, India. *7–11 Apr 2025*
8. *SKAO-Euclid Synergies of the post-Reionization Universe: 21cm-Galaxy Cross-Bispectrum*, **Cosmology with SKA and Beyond**, IIT Indore, India. *1–2 Apr 2025*
9. *Reionization via Higher-Order Statistics of Line Intensity Mapping*, **The Multi-Color Universe: A New Era with Line Intensity Mapping**, IWH, Heidelberg University, Germany. *2–5 Dec 2024*
10. *Impact of the Sources of Epoch of Reionization on the 21-cm Bispectrum*, **SKA Cosmic Dawn and Epoch of Reionization Science Team Meeting 2024**, Tsinghua University, Beijing, China (remote). *17–19 Jul 2024*
11. *Impact of the Sources of Epoch of Reionization on the 21-cm Bispectrum*, **ICTP Summer School on Cosmology**, ICTP Trieste, Italy. *17–28 Jun 2024*
12. *Impact of the Sources of Epoch of Reionization on the 21-cm Bispectrum*, **10th International Conference on Gravitation and Cosmology: New Horizons and Singularities in Gravity**, IIT Guwahati, India. *6–9 Dec 2023*
13. *A Numerical Investigation of the Inflationary Scenario in the Light of Planck*, **Seventh Southern Regional Astronomy Meeting: Research in Astronomy — Opportunities and Challenges**, (remote). *8–10 Sep 2021*

## SEMINARS

14. *Impact of the Sources of Epoch of Reionization on the 21-cm Bispectrum*, **Scuola Normale Superiore (SNS)**, Pisa, Italy. *9 Jul 2024*
15. *Impact of the Sources of Epoch of Reionization on the 21-cm Bispectrum*, **Institute for Fundamental Physics of the Universe (IFPU)**, Trieste, Italy. *1 Jul 2024*
16. *Impact of the sources of Epoch of Reionization on the 21-cm bispectrum*, **SKA India 21-cm CD/EoR Bi-weekly Meeting**, (remote), 15 Apr 2024.

## LIGHTNING TALKS AND POSTERS

17. *Probing the Large-scale Structure with 21cm-Galaxy Cross-bispectrum*, **New Frontiers in Cosmology 2026**, A Coruña, Spain (fireslide and poster). *10–14 Aug 2026*
18. *The 21cm-Galaxy Cross-Bispectrum: A Forecast for Synergies between SKA-Mid and Galaxy Surveys*, **SKA-India Science Meeting**, IISER Mohali, India (lightning talk and poster). *15–18 Jan 2026*
19. *21cm-CII cross-correlations: Improving the constraints on EoR source parameters via SKA*, **A New Era in Astrophysics: Preparing for Early Science with the SKAO**, Görlitz, Germany (lightning talk and poster). *16–20 Jun 2025*
20. *Impact of the Sources of Epoch of Reionization on the 21-cm Bispectrum*, **7th Global 21-cm Workshop**, RRI Bengaluru, India (lightning talk and poster). *30 Sep – 4 Oct 2024*
21. *Impact of the Sources of Epoch of Reionization on the 21-cm Bispectrum*, **Advanced 21-cm Cosmology Workshop**, NISER Bhubaneswar, India (poster). *18–21 Dec 2023*